

# Experiment Update

Real-only baseline and controlled brightness-synthetic experiment

## What happened after the first project review

The project completed a real-only baseline, trained a second model with 10% brightness-modified training images, and tested both models under normal, darker, and brighter validation conditions.

## Completed work

- Baseline v1 completed 15 epochs using only real Rural LoveDA images.
- The best real-only checkpoint occurred at epoch 10 with 24.42% mIoU and 54.52% pixel accuracy.
- A second, matched 15-epoch run added 152 brightness-modified copies to the training pool. Synthetic samples made up 10.01% of the final pool, while the number of training samples per epoch remained fixed.
- Both frozen models were evaluated on the same validation maps at five brightness levels. No labels were changed during these tests.

## Measured results

The table compares each model's best saved checkpoint. The only training difference was the controlled brightness synthetic data.

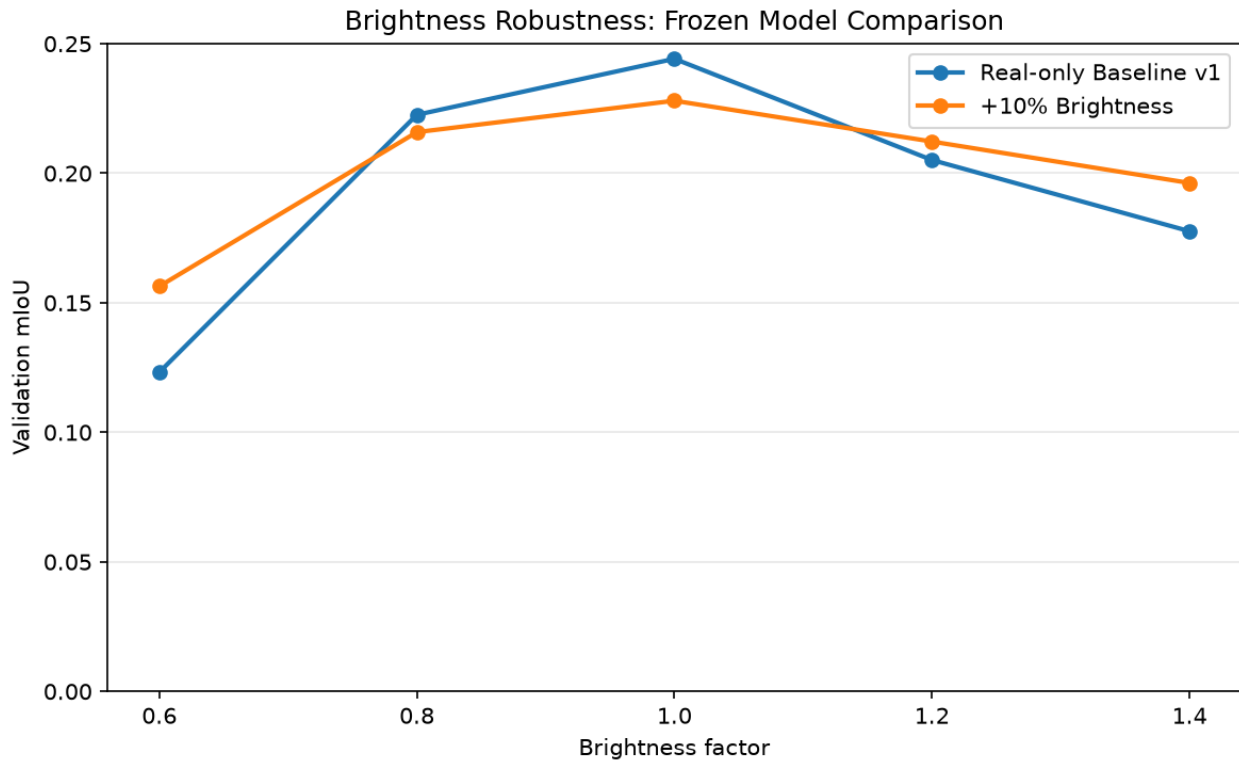
| Evaluation condition | Real-only mIoU | +10% Brightness mIoU | Difference   |
|----------------------|----------------|----------------------|--------------|
| Clean (1.0)          | 24.42%         | 22.80%               | -1.62 points |
| Darker (0.8)         | 22.26%         | 21.59%               | -0.67 points |
| Much darker (0.6)    | 12.32%         | 15.64%               | +3.32 points |
| Brighter (1.2)       | 20.51%         | 21.22%               | +0.71 points |
| Much brighter (1.4)  | 17.76%         | 19.62%               | +1.86 points |

## What this means

- On normal-looking validation imagery, the brightness-trained model performed worse. This is the clean-performance cost.
- On strong darkening and brightening, the brightness-trained model performed better. This is a robustness benefit.
- The result supports a nuanced conclusion: this synthetic data did not make the model universally better, but it did help it handle larger brightness shifts.
- This is one seed and one augmentation setting. Repeating the experiment with more seeds would make the conclusion stronger.

## Robustness evidence

The chart shows why evaluating only clean images would have missed the benefit under stronger brightness shifts.



## Example predictions

These five examples use the best brightness-trained checkpoint. Each row shows the original image, ground truth, prediction, and an error map in red.

